Koa Shen

Mechanical Engineering | Surgical Robotics, Mechanisms & Validation | Summer 2027

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EDUCATION

University of California, Santa Barbara Expected June 2028

B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94

- Dean's Honors every quarter. Relevant coursework: Dynamics, Mechanics of Materials, Circuits, MATLAB; Dynamical Systems and Machine Learning in progress.

ENGINEERING EXPERIENCE

Mechanical Engineering Intern Summer 2026 - Present

SLAC National Accelerator Laboratory, LCLS | Menlo Park, CA; extended part-time remote

- Built a Python/Drake digital twin for a 30-DOF, 305-body beamline assembly; millisecond collision checks identified at least five physical pinch points and informed mechanical redesigns.
- Created 15 reusable motion-stage models with CAD import, collision geometry, and visualization; the framework is being adopted across three LCLS assemblies.
- Redesigned optics packaging with kinematic mounts, alignment features, and detector strain relief; checked load cases to address crash and drop risks for $100k+ detectors.
- Produced Solid Edge designs and PDM-controlled documentation with GD&T for manufacture and inspection; path planning and controls integration remain in development.

Fabrication Team Lead Fall 2024 - Present

Gaucho Racing, Formula SAE Electric | UC Santa Barbara

- Led 15 teammates through driver safety-system design, manufacturing, and installation; GR26 passed FSAE technical inspection fully rules-compliant.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding critical bearing fits within 0.0005 in.
- Own chassis and ergonomics packaging, coordinating assembly interfaces, welded structures, and manufacturing across vehicle subteams.

Maintenance Technician Summer 2025

Exploratorium | San Francisco, CA

- Serviced 100+ public exhibits and completed three major overhauls; redesigned a load-spreading enclosure bezel to address acrylic cracking and avoid $3,000+ in replacement costs.
- Reverse-engineered mechanisms and redesigned encoder mounts for repeated use; documented CAD, CAM files, and remanufacturing instructions for future technicians.

SELECTED ROBOTICS & HARDWARE PROJECTS

Closed-Loop Cycloidal Actuator Summer 2026 - Present

- Designed and fabricated a 15:1 cycloidal reducer within a NEMA 17 motor's 42 mm square footprint, selecting PETG/ABS print orientations for load capacity and dimensional accuracy.
- Wrote PlatformIO C++ firmware for a TMC2209 driver and 12-bit AS5600 motor-shaft encoder; bench-validated position correction and missed-step detection on a single-joint prototype.

Formula SAE Steering-Wheel Development 2024 - 2026

- Helped reduce assembly mass from 5 lb to 1.7 lb through iterative design and FEA-guided optimization; integrated buttons, potentiometers, a display, and quick-release hardware.

TECHNICAL SKILLS

Mechanical Design: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, FEA, statics/dynamics, kinematics

Prototyping & Manufacturing: CNC/manual machining, Mastercam, welding, 3D printing, assembly, design for manufacturing

Robotics & Controls: Python, C/C++, MATLAB, Git, Drake, PlatformIO, Arduino, encoders, stepper control